

# Chapter 11 Summary — Coalition Formation and Detection (So Long Sucker)

Research on the possibilities for applying Artificial Intelligence in computer games

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## 11 Chapter 11 — Coalition Formation and Detection

This is a ground-up chapter on the thing that only exists once a game has a **third player**: temporary **alliances** that form, get exploited, and get betrayed. It uses **So Long Sucker (SLS)** — the four-player game of Hausner, Nash, Shapley & Shubik, published in 1964, whose outcome depends almost entirely on bargaining between the players<sup>1</sup> — and extends the only published RL treatment of it<sup>2</sup>, whose agents are coalition-blind, with a coalition detector and a Shapley-based reward. Through it the chapter internalizes the jump from  $N = 2$  to  $N \geq 3$ , where **Nash and exploitability stop being tractable *and* stop being meaningful**, so “did it work?” can no longer be a single number. It is written to be read on its own. **All experimental numbers reported here were measured** on reproducible runs and, wherever possible, are bounded by *exact* references (textbook Shapley/core values for the cooperative-game toys; an exact 2-player minimax solver for the SLS endgame). Where a run contradicted what I expected — including a real engine bug — I keep the original expectation and reconcile it with what happened; those gaps are the most instructive parts of the chapter.

**Where this sits in the thesis.** Chapters 2-10 all leaned on a 2-player game with an *exact* best-response oracle. Chapter 11 removes it and enters the frontier. Three thesis hooks live here. The **coalition detector** lifts opponent modeling from “what kind of player is this?” to “who is allied with whom?”

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<sup>1</sup>Hausner, M., Nash, J., Shapley, L. & Shubik, M. (1964). “So Long Sucker - A Four-Person Game.” In M. Shubik (Ed.), *Game Theory and Related Approaches to Social Behavior*, 359–361. New York: Wiley.

<sup>2</sup>Sharan, M. & Adak, C. (2024). “Reinforcing Competitive Multi-Agents for Playing ‘So Long Sucker’” *arXiv:2411.11057*.

(Contribution #1). The **safe baseline loses its Nash anchor** — with no minimax value and an empty core there is no stable allocation for safety to rest on; regularizing toward a behavioral anchor in the spirit of piKL is a candidate but gives no guarantee — the gap this chapter frames but does not close (Contribution #2). And the **EGTA meta-game + Shapley credit** stand in for exploitability, which with more than two players no longer measures what an agent is guaranteed, least of all against a coalition (Contribution #3).

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## 11.1 Why the third player changes everything

With two players, a zero-sum game has a *value*: there is a minimax-optimal strategy, and “how far from optimal are you?” (exploitability) is a single, meaningful number that anchored every chapter since Chapter 2. Add a third player and the ground shifts. Now two players can **gang up** on the third, and the interesting question is not “what is the equilibrium?” but “who allies with whom, for how long, and who betrays first?” Nash equilibrium in a 4-player free-for-all is both **intractable** to compute and **without guarantees**: finding one is at least as hard as in general two-player games<sup>3</sup>, and equilibria chosen independently need not form an equilibrium<sup>4</sup>; it also guards only against unilateral deviations, not against the coalitions that actually decide the game<sup>5</sup>.

A picture to hold onto: a **dinner party** with four guests and one house. Everyone pairs up to get things done, but the smartest move is to have quietly betrayed your partner one turn *before* they betray you. So Long Sucker makes this literal: you hold coloured chips, you place them into other players’ piles (a handshake) or capture their piles (the knife), and you are eliminated when you run out. In this chapter’s engine the alliance is **encoded entirely in the moves**: the free, unenforceable negotiation of the real game is not modelled, which is why a detector can read the coalition straight off the move stream.

The consequence for methodology is the single most important idea in the chapter: **at  $N \geq 3$  you trade exact evaluation for empirical evaluation**. No exploitability; instead win-rate, a coalition score from the detector, and the cyclic ratio of an empirical meta-game — all anchored by the one subgame that *is* exactly solvable, the 2-player endgame.

**Read more:** Hausner, Nash, Shapley & Shubik (1964) — the rules of the game; De Carufel & Jerade (2024) — a characterization of the two-player endgame<sup>6</sup>. The exact anchor in this chapter is the engine’s own minimax solution, not yet cross-checked against that characterization.

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<sup>3</sup>Daskalakis, C., Goldberg, P. W. & Papadimitriou, C. H. (2009). “The Complexity of Computing a Nash Equilibrium.” *SIAM Journal on Computing*, 39(1), 195–259.

<sup>4</sup>Brown, N. & Sandholm, T. (2019). “Superhuman AI for multiplayer poker.” *Science*, 365(6456), 885–890.

<sup>5</sup>Bernheim, B. D., Peleg, B. & Whinston, M. D. (1987). “Coalition-Proof Nash Equilibria I. Concepts.” *Journal of Economic Theory* 42(1), 1–12. DOI 10.1016/0022-0531(87)90099-8.

<sup>6</sup>De Carufel, J.-L. & Jerade, M. R. (2024). “So Long Sucker: Endgame Analysis.” *arXiv:2403.17302*.

## 11.2 The coalition detector — reading alliances off the moves

The first tool generalizes opponent modeling. Instead of inferring a hidden *type* or *hand* (Chapter 7), the detector infers a hidden *social structure*. It watches the move log and accumulates two matrices: **help** (player  $i$  placed a chip into player  $j$ 's pile) and **harm** (player  $i$  captured player  $j$ 's pile). Their difference is **net support**; the pair with the highest mutual net support is the strongest coalition.

In this engine the alliance is visible only in the moves - negotiation is not modelled.

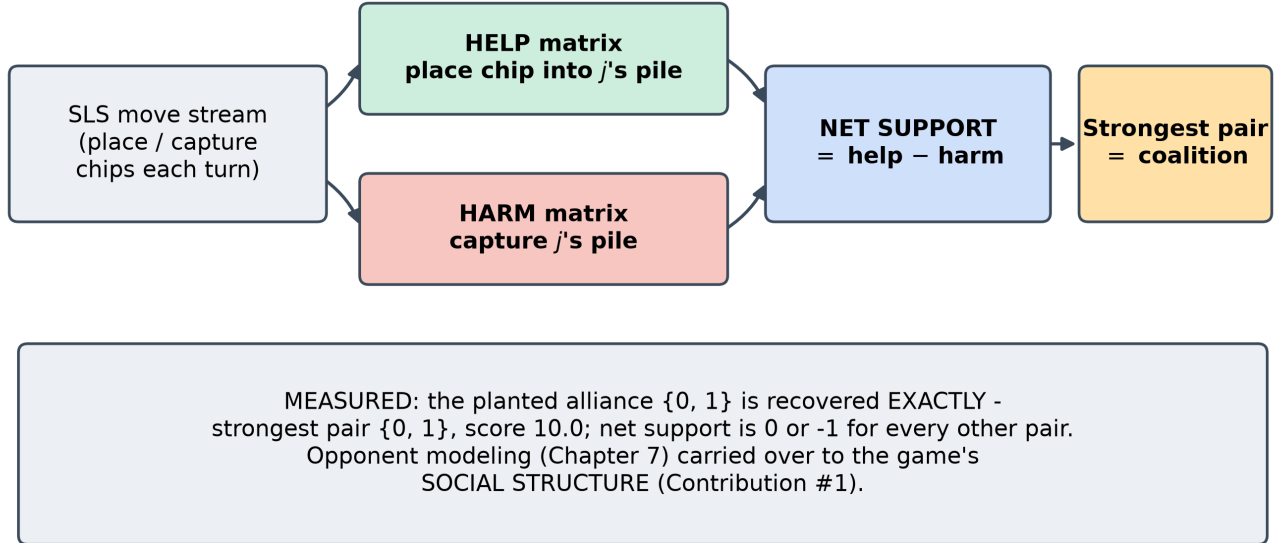


Figure 1: The coalition detector: from the SLS move stream, placing a chip into another player's pile counts as **HELP** and capturing a pile counts as **HARM**. Accumulated into help/harm matrices and differenced into net support, a reciprocal alliance shows up as a strong mutual edge. Opponent modeling (Chapter 7) lifted from “what hand?” to “who is allied with whom?” (Contribution #1).

Does it work? We script two players to systematically help each other and harm the rest, tell the detector nothing, and ask it to name the coalition. Measured, it recovers the planted  $\{0, 1\}$  alliance **exactly**, with a well-separated score (10.0 on the allied pair, 0 or  $-1$  everywhere else). The alliance is fully legible from chip placement alone. This is a sanity check against one scripted, overt pair; false positives on non-colluding play and covert colluders were not tested.

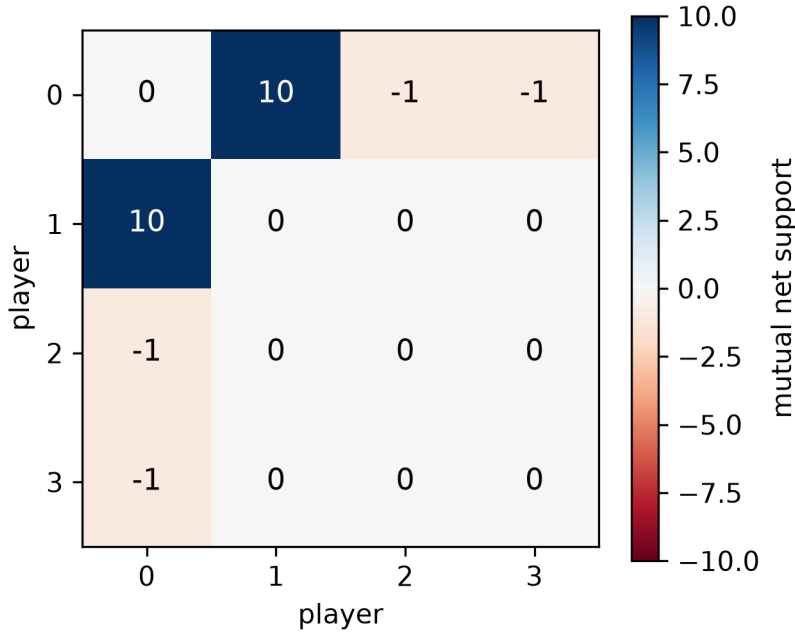


Figure 2: Coalition-score matrix inferred purely from chip placement: the planted alliance between players 0 and 1 shows as strong mutual support (10), while the other pairs are neutral (0) or hostile (-1). The detector recovers a coalition it was never told about.

**Read more:** the Chapter 7 opponent-modeling implementation — the “observe actions -> update beliefs” principle the detector re-derives as help/harm matrices.

### 11.3 Shapley credit for a competitive game

If we want an agent to *learn* to form coalitions, we need a reward signal that says “you contributed to a coalition.” The classical tool is the **Shapley value**: the unique split of a coalition’s worth that satisfies Shapley’s axioms (efficiency, symmetry, null player, additivity), averaging each member’s marginal contribution over all join orders. Two textbook toys pin down what “fair” and “stable” mean, and both reproduce exactly:

Table 1: Shapley value and core stability on two cooperative-game toys.

| Cooperative-game toy | Shapley value     | Core (stability)                         |
|----------------------|-------------------|------------------------------------------|
| Glove game           | $(2/3, 1/6, 1/6)$ | <b>non-empty</b> (a stable split exists) |
| 3-player majority    | $(1/3, 1/3, 1/3)$ | <b>empty</b> (no stable split)           |

The **empty core** of the majority game is the conceptual heart of the chapter: it is a game where *any*

allocation can be overturned by some coalition, so cooperation is *inherently unstable*. SLS is constant-sum (only one player wins), and every essential constant-sum game has an empty core<sup>7</sup>; so the SLS coalition game, in which a coalition’s value is the win probability it can guarantee, has no stable allocation either (it is essential because no player can guarantee a win alone). Together with the absence of a minimax value beyond two players, this is why N-player “safe” play cannot be anchored to a stable equilibrium (Contribution #2).

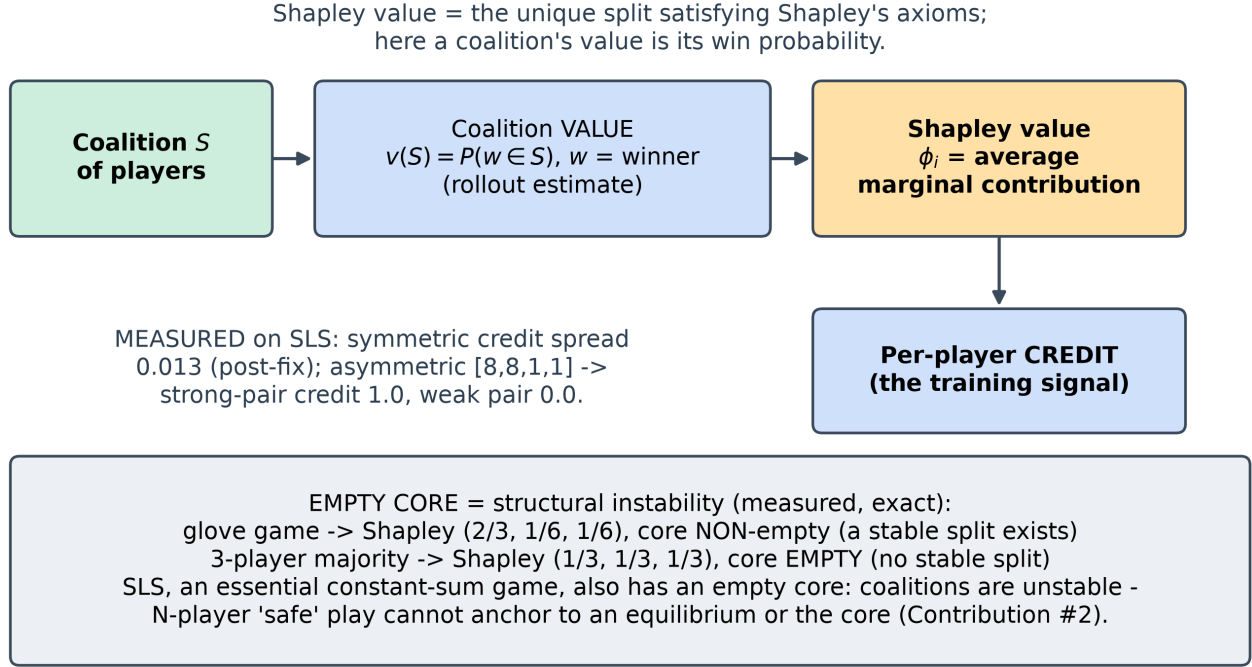


Figure 3: Shapley credit adapted to a purely competitive game: the coalition’s “value” is redefined as the PROBABILITY a coalition member wins (estimated by rollouts), and each player’s Shapley value of that win-probability function is their credit. The empty core of the majority game (no stable allocation) is the structural signature of SLS coalitions - they will break.

SLS is competitive, not cooperative — there is no shared pot to split — so we **redefine the coalition value** as *the probability that a member of the coalition wins*, estimated by Monte-Carlo rollouts. Each player’s credit is then the Shapley value of that win-probability function. Measured on SLS positions: a genuinely symmetric position gives near-equal credit (spread 0.013), and an asymmetric [8, 8, 1, 1] position hands *all* credit to the strong pair (coalition value 1.0). The symmetric result is reported **after** a bug fix — see the reconciliation below.<sup>8</sup>

<sup>7</sup>Ferguson, T. S. *Game Theory* (lecture notes, UCLA), Part IV “Games in Coalitional Form”, § 2.3, Theorem 1.

<sup>8</sup>Shapley, L. S. (1953). “A value for n-person games.” In *Contributions to the Theory of Games II* (Annals of Mathematics Studies 28), 307–318. Princeton University Press; Chalkiadakis, G., Elkind, E. & Wooldridge, M. (2011). *Computational Aspects of Cooperative Game Theory*. Morgan & Claypool (core, Shapley value, nucleolus); Wang, J., Zhang, Y., Kim, T.-K. & Gu, Y. (2020). “Shapley Q-Value: A Local Reward Approach to Solve Global Reward Games.” *AAAI* 34(5), 7285–7292 (Shapley-based credit assignment in MARL).

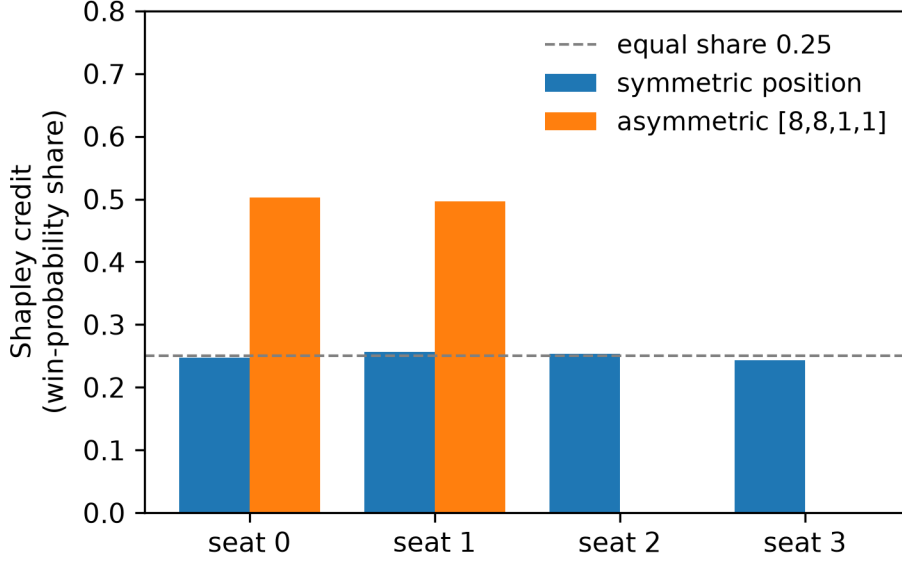


Figure 4: Shapley credit on two SLS positions: near-equal across the four seats in the symmetric position (spread 0.013 after the fix) and fully concentrated on the strong pair in the asymmetric position [8, 8, 1, 1]. Credit follows the real advantage only once the engine’s tie-break rule is unbiased.

**Reconciliation (kept prediction -> what actually happened).** I predicted a symmetric position would give a symmetric credit spread ( $< 0.15$ ). The first run gave 0.54 — a red FAIL — with Player 0 winning  $\sim 2\times$  its fair share across three independent scripts. Suspecting the engine before the prediction, I found the mechanism:  **$\sim 99.5\%$  of random games in this engine end in a deadlock** (all live hands empty — a consequence of its simplified rules, which skip a player with an empty hand), so the winner is decided by a most-chips **tie-break** — whose lowest-index rule quietly handed seat 0 its edge. An **unbiased random tie-break** fixed it: symmetric spread  $0.54 \rightarrow 0.013$ , all-random winners now uniform. It also revealed that an impressive  $\sim 0.87$  hero win-rate had been the *same* artifact (the hero always sat in seat 0); the fair number is  $\sim 0.41$ . The lesson: in a game that almost always ends in a near-tie, the tie-break rule is the most load-bearing line in the engine, and a symmetric *position* is not a symmetric *outcome* until it is unbiased.

#### 11.4 Coalition-aware MAPPO — and when the coalition score actually rises

Now we train. Each SLS seat is a masked, episodic PPO agent learning by self-play. The reward is a **blend**:

$$r = \alpha r_{\text{sparse}} + (1 - \alpha) (\text{Shapley coalition credit}),$$

where  $r_{\text{sparse}}$  is the sparse winner-takes-all reward and the second term is the centered Shapley credit from Section 11.3, computed either cheaply from critic values (**proxy**) or by rollouts (**counterfactual** in the

code). The weight  $\alpha$  dials between “just win” ( $\alpha = 1$ ) and “form coalitions” ( $\alpha = 0$ ).

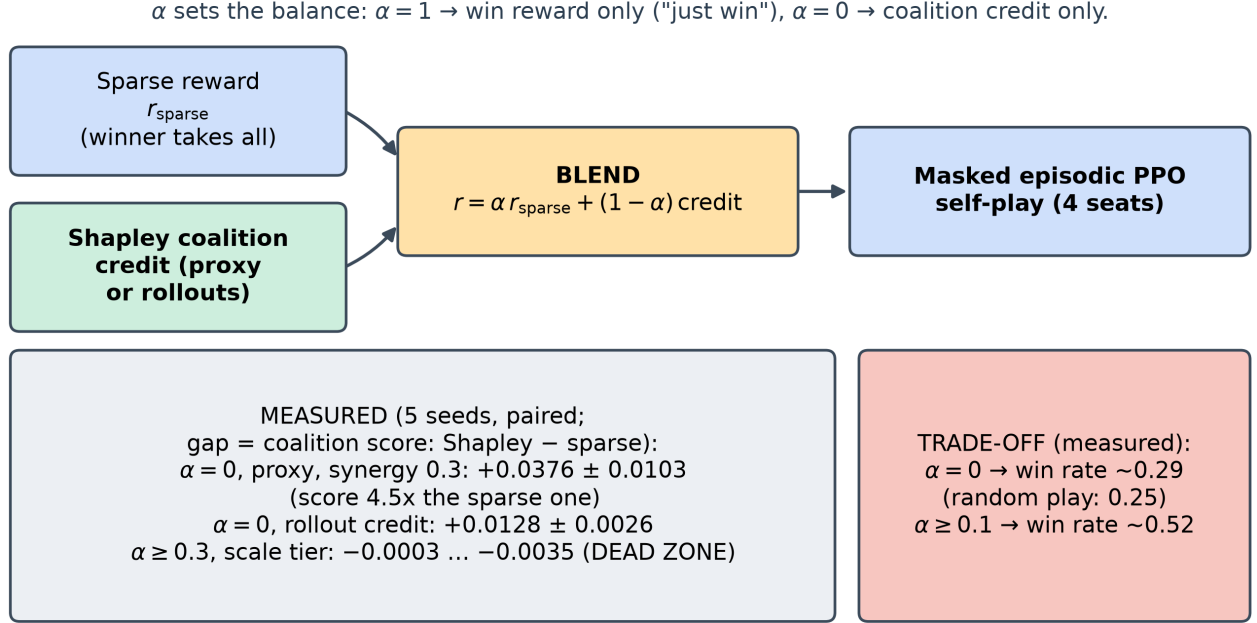


Figure 5: Coalition-aware MAPPO reward blend: the sparse winner-takes-all signal is mixed with the Shapley coalition credit by weight alpha (alpha = 1: sparse only, alpha = 0: credit only). Measured, alpha is the dominant knob: the coalition score rises significantly only at low alpha, and every alpha  $\geq 0.3$  suppresses the signal. The cheap critic-value proxy has a larger effect than the rollout credit.

Do coalition-aware agents form coalitions more than sparse agents? The honest answer required a **5-seed paired sweep** over  $\alpha \times$  credit-mode  $\times$  synergy, because a single config misled me badly (below). There are two tiers: smoke (5 chips, 400 training games) and scale (7 chips, 1500 games). Measured (paired gap = coalition score of Shapley agents minus sparse agents; \*\* = gap above two standard errors, which with 5 seeds corresponds to roughly  $p < 0.12$ ):

Table 2: Paired coalition-score gap of Shapley credit over the sparse baseline, per regime.

| Regime                                   | Paired gap (Shapley - sparse)                                                           |
|------------------------------------------|-----------------------------------------------------------------------------------------|
| scale, proxy, $\alpha = 0$ , synergy 0.3 | $+0.0376 \pm 0.0103$ (Shapley agents’ score $\sim 4.5$ x the sparse baseline’s)         |
| scale, rollout credit, $\alpha = 0$      | $+0.0128 \pm 0.0026$                                                                    |
| scale, $\alpha \geq 0.3$                 | $-0.0003 \dots -0.0035$ (all negative; smoke: $-0.003 \dots +0.001$ , none significant) |
| smoke, proxy, $\alpha = 0$ , synergy 0.1 | $+0.0024 \pm 0.0008$ (tiny)                                                             |

**Reconciliation (kept prediction -> what actually happened).** My single-config runs used the default  $\alpha = 0.3$  and showed the coalition signal collapse at scale — which I first read as “the proxy credit is too weak once training is longer.” The sweep overturned that completely:  **$\alpha$  is the dominant knob, and 0.3 is a dead zone.** The coalition score rises significantly *only at low  $\alpha$*  (at  $\alpha \approx 0$  the Shapley agents exceed sparse by +0.038), while *every* scale-tier  $\alpha \geq 0.3$  cell is negative — the sparse winner-takes-all term drowns the coalition signal. Two further surprises: the effect is **larger in the scale tier** (7 chips and 1500 training games vs 5 and 400, so game size and training length are confounded; opposite to my “smoke-positive / scale-null” read, which was an artifact of holding  $\alpha = 0.3$  at both tiers), and the **cheap proxy beats the expensive rollout credit**. The effect is robust at low  $\alpha$  — I had simply measured it in the one regime where the sparse term hides it; whether it reflects coalition formation is not yet tested (see the caveat below). The fix is “weight the coalition credit heavily,” not “compute a truer credit.”

**Mechanism caveat.** In `shapley.py` the rollout value is  $v(S) = \sum_{i \in S} P(i \text{ wins})$  — an additive game, in which each player’s Shapley value is simply its own win probability (and in training it is computed once per batch, from the initial state). The centered proxy credit is proportional to the agent’s own critic estimate minus the table mean; synergy only rescales it. Neither signal encodes who helped whom, and the coalition score is the mean *absolute* mutual net support, so it also rises with mutual hostility. The low- $\alpha$  effect therefore shows that replacing the win reward with a dense signal changes how the agents interact; whether coalitions form that way needs a control with an equally dense reward that carries no coalition information.

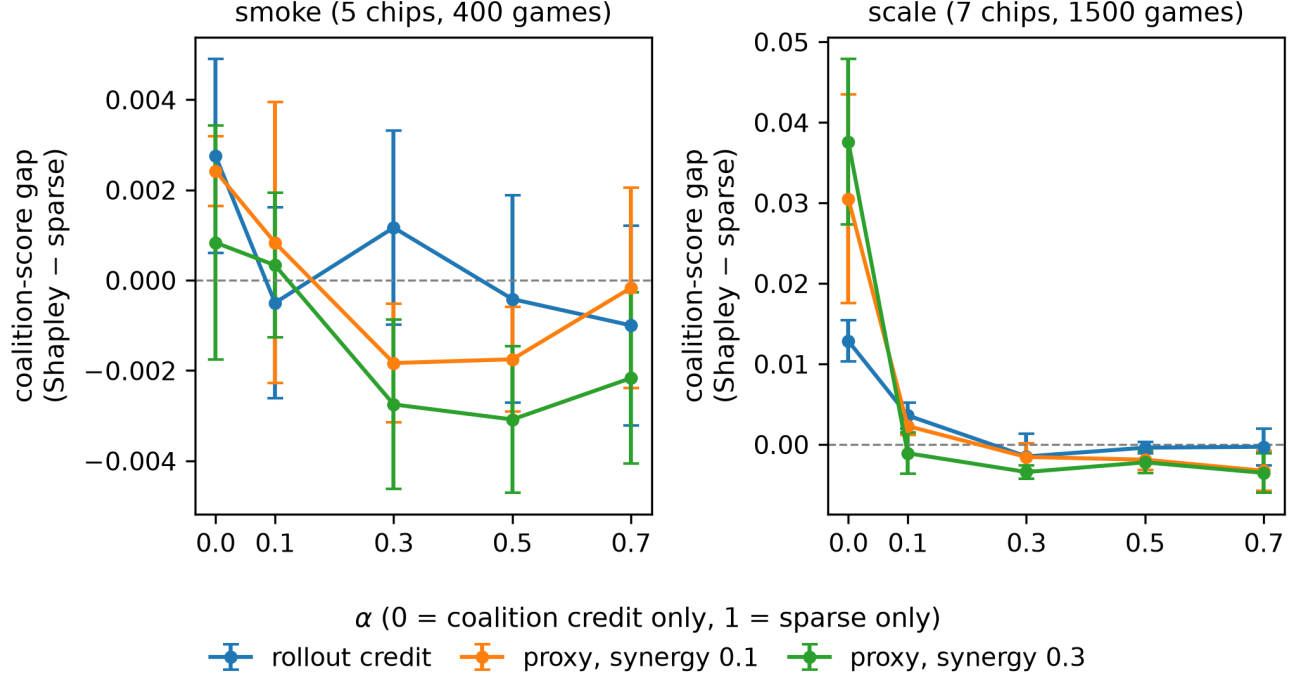


Figure 6: Paired coalition-score gap across the alpha x credit x synergy sweep (5 seeds, error bars = 1 SE): large and significant only in the low-alpha regime (peaking at +0.038 with the proxy at alpha=0); in the scale tier it is negative for every alpha  $\geq 0.3$ . The earlier null came from measuring in the alpha=0.3 dead zone.

There is no free lunch: pure coalition credit ( $\alpha = 0$ ) drops win-rate to  $\sim 0.29$  (near the 0.25 random floor), while  $\alpha \geq 0.1$  keeps it  $\sim 0.52$ . Coalition-*forming* is the primary target and winning is secondary — exactly as the chapter’s plan framed it, now quantified.<sup>9</sup>

## 11.5 EGTA and the spinning top — is SLS a wheel or a ladder?

The final tool evaluates the *population*. **Empirical game-theoretic analysis (EGTA)** treats whole strategies as the atoms of a meta-game, plays every pair to fill an empirical payoff matrix, and analyzes its structure. To reuse Chapter 9’s meta-Nash solver and Chapter 10’s **spinning-top** (Hodge) decomposition — both 2-player tools — the 4-player payoff *tensor* is **projected** to a pairwise matchup matrix, then split into a **transitive** (skill-ladder) component and a **cyclic** (rock-paper-scissors) component.

<sup>9</sup>The MAPPO implementation with a centralized critic is from Chapter 9. piKL (regularizing search toward a human-like policy) was introduced by Jacob, A. P. et al. (2022). “Modeling Strong and Human-Like Gameplay with KL-Regularized Search.” *ICML*, arXiv:2112.07544, and applied to Diplomacy by Bakhtin, A. et al. (2023). “Mastering the Game of No-Press Diplomacy via Human-Regularized Reinforcement Learning and Planning.” *ICLR*, arXiv:2210.05492. The anchor there imitates human play and carries no safety guarantee; using it as the anchor for N-player safe exploitation is this thesis’s proposal.

EGTA = an empirical meta-game whose "strategies" are whole policies;  
exploitability gives no guarantee against a coalition.

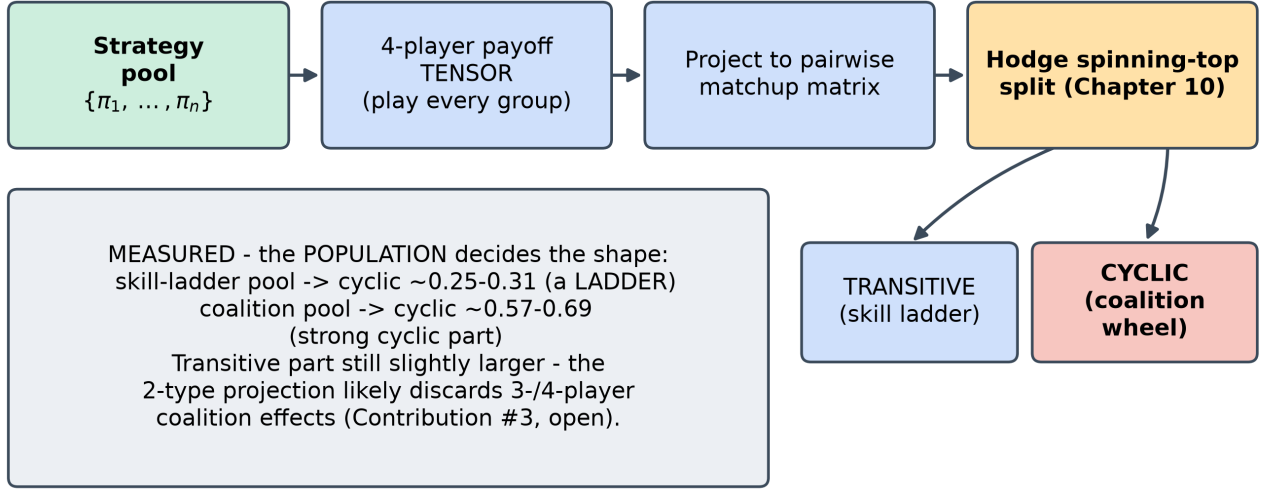


Figure 7: The EGTA + spinning-top pipeline for SLS: play every pair of strategies to fill a payoff tensor, project the 4-player tensor to a pairwise matchup matrix, and Hodge-decompose it into transitive (skill ladder) and cyclic (coalition counters) parts. Measured caveat: the 2-type projection likely discards 3-/4-player coalition effects, so the cyclic ratio may be underestimated.

Chapter 10 predicted FFA coalition games would be strongly cyclic. Measured, it depends entirely on **which population you decompose** — the same lesson Chapter 10 taught:<sup>10</sup>

Table 3: Cyclic ratio and structure of two SLS populations.

| Population                              | Cyclic ratio | Structure                                                    |
|-----------------------------------------|--------------|--------------------------------------------------------------|
| Skill-ladder pool                       | 0.25-0.31    | transitive-dominant (a ladder)                               |
| Coalition pool (ally/betray strategies) | ~ 0.57-0.69  | strong cyclic component;<br>transitive still slightly larger |

**Reconciliation (kept prediction -> what actually happened).** I expected a large cyclic component and, at first, saw a near-perfect skill ladder (cyclic  $\sim 0.07$ ). That was partly the seat-0 bug (Section 11.3) and partly **pool composition**: the default baseline pool *is* a skill ladder. A coalition pool (fixed-ally + betrayer strategies) pushes the cyclic ratio to  $\sim 0.57$ -0.69 — a large non-transitive component that strongly confirms the *direction* of the prediction, but stays **honestly short of strict dominance** (cyclic<sup>2</sup> just under 0.5). The prime suspect for

<sup>10</sup>Balduzzi, D. et al. (2019). “Open-ended Learning in Symmetric Zero-sum Games.” *ICML*, PMLR 97, 434–443 — the transitive/cyclic decomposition; Czarnecki, W. M. et al. (2020). “Real World Games Look Like Spinning Tops.” *NeurIPS*, arXiv:2004.09468 — the spinning-top geometry.

the residual is that the **2-type projection discards 3-/4-player coalition effects** — a tensor-native decomposition is the open question. Neither reading is a bug; which population you build decides whether SLS looks like a wheel or a ladder.

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## 11.6 Honest notes, limitations, and where this hands off

**What held up.** The exact anchors are solid: the SLS engine matches the 2-player minimax endgame with **zero** mismatches; the detector recovers a planted coalition **exactly** ( $\{0, 1\}$ , score 10.0); the Shapley code reproduces the glove and majority toys — *including their core* — to four decimals. The headline training result holds in a 5-seed paired sweep: at low  $\alpha$  the coalition score is significantly higher (+0.038), although, as implemented, the Shapley credit carries no coalition information (Section 11.4).

**What did not, and why it matters.** Two honest corrections travel forward. (1) The “symmetric” game was not symmetric — a most-chips **tie-break bug** handed seat 0  $\sim 2\times$  its fair share and inflated the hero win-rate from a true  $\sim 0.41$  to a false  $\sim 0.87$ ; fixed, but it proves the engine’s *rules*, not its solver, are where the risk lives. (2) “Coalitions don’t emerge at scale” was a **mis-set blend weight** ( $\alpha = 0.3$  is a dead zone), not a failure of the method. A methodological echo of Chapters 9-10: **a single config hides what a seeded sweep reveals**.

**Trust.** Every exact target (endgame minimax, cooperative-GT toys) is deterministic and reproducible; the training claims rest on a 5-seed paired sweep with error bars, so the *directions* are the trustworthy claims, not third-decimal magnitudes. The standing caveat is **engine fidelity**: the engine is certified against its own ruleset, not against the De Carufel & Jerade formalization — and the tie-break bug shows that reconciliation is not cosmetic. (The committed `scale_results.json` is a pre-fix artifact, cited only as evidence of the bug; authoritative scale numbers come from the sweep.)

**Connections to other chapters.** Backward: the detector is Chapter 7’s opponent model lifted to social structure; Chapters 9-10’s meta-Nash and spinning-top decomposition are reused on the projected meta-game; with no minimax value and an empty core, Chapter 8’s definition “safe = bounded deviation from Nash” loses its footing, and regularizing toward a behavioral anchor (in the spirit of piKL) remains a candidate without a guarantee. Forward: Chapter 12 treats language models as strategic agents but not negotiation; N-player safety (Contribution #2) and EGTA evaluation on the payoff tensor as a multi-agent generalization of exploitability (Contribution #3) remain tasks for the thesis.

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## 11.7 Key takeaways for the thesis synthesis

- **At  $N \geq 3$ , exact evaluation is gone.** Nash is intractable and offers no guarantee in 4-player FFA; evaluation becomes empirical (win-rate + coalition score + EGTA cyclic ratio), anchored by the one exactly-solvable subgame (the 2-player minimax endgame, 0 mismatches).
- **Replacing the win reward with a dense credit raises the coalition score, which is not**

**yet evidence of learned coalitions.** At  $\alpha \approx 0$  the Shapley-credit agents' coalition score exceeds the sparse agents' by +0.038 (5 seeds, scale tier); at  $\alpha \geq 0.3$  the effect vanishes. As implemented, the credit does not encode who helped whom (the proxy reduces to each agent's own critic value minus the table mean, the rollout variant to its own win-probability share), and the coalition score also counts hostile interactions; a coalition-free dense-reward control is needed.

- **Weighting the credit costs winning.**  $\alpha = 0$  drops win-rate to  $\sim 0.29$ , near the 0.25 random level;  $\alpha \geq 0.1$  restores  $\sim 0.52$  — the price of the dense signal, now measured.
- **Empty core  $\Rightarrow$  structural instability.** As a constant-sum game, SLS has an empty core, like the majority game: no stable allocation exists, so coalitions *will* break. N-player “safe” play therefore cannot be anchored to a Nash equilibrium or the core; a behavioral anchor in the spirit of piKL is a candidate, not a guarantee (Contribution #2).
- **Which population you decompose decides ladder-vs-wheel** (the Chapter 10 lesson, confirmed): a skill-ladder pool looks transitive ( $\sim 0.3$  cyclic), a coalition pool has a strong cyclic component ( $\sim 0.57$ - $0.69$ ), though the transitive part stays slightly larger.
- **Engine rule fidelity outranks solver code.** Two of the remaining weaknesses (the seat bias, the sub-dominant cyclic ratio) live in the engine's simplifications, not in the Shapley/EGTA math, which passed its textbook checks exactly; the third, the coalition-free credit (Section 11.4), is in how the credit is defined.